Assignment 1: Robot Localization in Simulation

Intelligent Robotics — Individual Assignment

1. Problem Statement

This assignment addresses the localization of a differential-drive mobile robot in a 2D simulated indoor environmen

2. Methods

Simulation Platform: ROS 2 Jazzy with Gazebo Harmonic; differential-drive vehicle_blue model.

Localization Method: Dead reckoning with the Euler-integrated motion model:

$$x(k+1) = x(k) + v\cos(\theta(k))\,dt$$
$$y(k+1) = y(k) + v\sin(\theta(k))\,dt$$
$$\theta(k+1) = \theta(k) + \omega\,dt$$

with dt = 0.1 s. Uncertainty is modelled by adding Gaussian noise to the commanded velocities: v_odom = v_cmd

Ground Truth: The simulator's true pose is obtained from /model/vehicle_blue/odometry and is used only for evalua

3. Results

The estimated trajectory closely follows the ground-truth trajectory throughout the experiment. The final position e

Error accumulates gradually due to the integration of noisy velocity measurements. The straight-line segments sho

Sources of error include: (1) Gaussian noise added to the velocity commands, (2) discretization error from Euler int

The result demonstrates that dead reckoning provides reasonable short-term localization, but drift will grow unbou

Trajectory plot: results/trajectory_plot.pdf